

INTRODUCTION TO COMPUTING

EASY REVIEWER

Based only on the uploaded lessons: Introduction to Computers, Hardware, and Data Representation.

PART 1 — INTRODUCTION TO COMPUTERS

What is a computer?

A computer is an electronic device that can store, retrieve, and process data. It can do simple calculations and complex tasks.

Characteristics of Computers

Characteristic	Easy meaning
Speed	Can perform millions of operations in a very short time.
Storage Capacity	Can store a large amount of information.
Accuracy	Can give accurate results when the data and instructions are correct.
Diligence	Does not get tired or bored when doing repeated tasks.
Versatility	Can perform many different kinds of tasks.
Automation	Can perform tasks automatically when properly programmed.
Reliability	Can perform tasks consistently and repeatedly.
Connectivity	Can connect to the internet and other devices to share information.
Multitasking	Can handle several tasks at the same time.
No Intelligence of Their Own	Follows human-given instructions and cannot make decisions like humans unless programmed to do so.

Classification and Types of Computers

Supercomputer — Most powerful type; used for weather forecasting, scientific research, and complex calculations.

Mainframe Computer — Large computer used by banks, government offices, and large organizations to process huge amounts of data.

Minicomputer — Medium-sized computer that can support several users at the same time.

Microcomputer — Smaller computer mainly for individual users, such as desktops and laptops.

General-Purpose Computer — Used for many tasks such as browsing, documents, videos, and games.

Special-Purpose Computer — Designed for a specific task, such as ATMs, traffic lights, and factory machines.

Analog Computer — Processes continuous data such as temperature, speed, and pressure.

Digital Computer — Processes data using numbers and symbols; most modern computers and smartphones are digital.

Hybrid Computer — Combines analog and digital features; used in medical and scientific equipment.

Advantages of Computers

- Fast — processes large amounts of information quickly.
- Accurate — gives accurate results when correct data and instructions are used.
- Easy Communication — supports email, video calls, and social media.

- Information Access — helps users find information quickly through the internet.
- Storage — stores many files, pictures, videos, and documents.
- Helps in Education — useful for research, online classes, presentations, and assignments.
- Saves Time and Effort — performs repetitive tasks faster than people.

Disadvantages of Computers

- Health Problems — long use can cause eye strain, back pain, and less physical activity.
- Cybersecurity Risks — computers can be affected by viruses, hacking, and other online threats.
- Expensive — buying and maintaining computers can cost a lot.
- Addiction — too much gaming, social media, or internet use can become excessive.
- Job Replacement — automation can replace or reduce some jobs.
- Privacy Problems — personal information can be stolen or misused online.
- Dependence on Technology — people may become too dependent on computers for everyday tasks.

Applications of Computers

Education — Research, online learning, presentations, and school projects.

Business — Records, reports, customer communication, and financial transactions.

Banking — Transactions, accounts, money transfers, and online banking.

Medicine — Patient records, medical tests, assistance to doctors, and patient monitoring.

Communication — Email, messaging, social media, and video conferencing.

Entertainment — Movies, music, games, and digital content.

Science and Research — Data analysis, calculations, simulations, and research.

Government — Records, information management, online services, and documents.

Engineering and Architecture — Designs, plans, models, and simulations.

Transportation — Airplanes, trains, ships, traffic systems, and navigation systems.

PART 2 — COMPUTER HARDWARE

Hardware means the physical, tangible parts of a computer or electronic system — the parts you can touch. It is different from software, which is intangible code and instructions.

Three Main Groups of Hardware

- 1. **Processor and Memory** — The CPU executes instructions. RAM/ROM hold data and instructions needed by the computer.
- 2. **Secondary Storage** — Keeps data and programs permanently, even when the computer is switched off.
- 3. **Input/Output Devices** — Input devices put data into the computer; output devices give results back to the user.

Processor and Memory

CPU (Central Processing Unit) — Executes instructions and processes the work of the computer.

RAM (Random Access Memory) — Temporary, fast working memory. It holds data and instructions the CPU is currently using. Its contents are cleared when power is off.

ROM (Read-Only Memory) — Permanent memory that stores startup instructions needed to boot the computer. Its contents are not lost when power is turned off.

Secondary Storage

HDD (Hard Disk Drive) — Uses magnetic spinning disks; large capacity at low cost and common for bulk storage.

USB Flash Drive — Small, portable flash storage used to move files between computers.

SSD (Solid State Drive) — Uses flash memory with no moving parts and has faster read/write speeds than HDD.

Optical Disc (CD/DVD) — Uses a laser to read and write data; used for media and software distribution.

Input and Output Devices

Type	Device and purpose
Input	Keyboard — types text and commands.
Input	Mouse — points, clicks, and selects.
Input	Scanner/Camera — captures images.
Input	Microphone — captures sound.
Output	Monitor — displays visual output.
Output	Printer — produces output on paper.
Output	Speakers — produce sound output.

PART 3 — DATA REPRESENTATION

Data Representation explains how data is transformed into a coded form so computers can store and process it. Digital computers represent data using binary-coded forms.

Number Systems

A number system is a system used to represent numbers. The four main number systems in the lesson are:

Number system	Base	Digits / easy meaning
Binary	Base 2	Uses 0 and 1.
Octal	Base 8	Uses digits 0 to 7.
Decimal	Base 10	Uses digits 0 to 9; the usual number system used in everyday life.
Hexadecimal	Base 16	Uses 0–9 and A–F. A–F represent decimal 10–15.

Important: 8 bits = 1 byte. Binary uses only 0 and 1.

Number System Conversion

- Binary → Octal: group binary digits into sets of 3, then replace each group with its octal digit.
- Binary → Hexadecimal: group binary digits into sets of 4, then replace each group with its hexadecimal digit.
- Octal → Binary: replace each octal digit with its corresponding 3-bit binary string.
- Hexadecimal → Binary: replace each hexadecimal digit with its corresponding 4-bit binary string.

Example from the lesson: 10001011■ → 010 001 011 → 213■.

Example from the lesson: 10001011■ → 1000 1011 → 8B■■■.

Binary Arithmetic

Binary arithmetic is used in digital computers because numbers are stored in binary form. The four operations are addition, subtraction, multiplication, and division.

Binary Addition

- $0 + 0 = 0$
- $1 + 0 = 1$
- $1 + 1 = 10$ ■ (0 with carry 1)
- $1 + 1 + 1$ (carry) = 11 ■ (1 with carry 1)

Binary Subtraction

- $0 - 0 = 0$
- $1 - 0 = 1$
- $1 - 1 = 0$
- $0 - 1$: borrow 1 from the next column, making $10 - 1 = 1$.

Binary Multiplication

- $0 \times 0 = 0$
- $1 \times 0 = 0$
- $1 \times 1 = 1$

Binary Division

1. Start with the working portion of the dividend by including digits until the divisor can go into it.
2. Determine how many times the divisor goes into the working portion. In binary, this is 1.
3. Subtract the divisor from the working number.
4. Bring down digits until the new working number is large enough.
5. Repeat until the end of the dividend.
6. The result of the final subtraction is the remainder.

QUICK MEMORY GUIDE

- Computer = stores + retrieves + processes data.
- Hardware = parts you can touch.
- CPU = executes instructions.
- RAM = temporary memory; power off clears it.
- ROM = permanent startup memory.
- HDD / SSD / USB / CD-DVD = secondary storage.
- Input = data goes IN (keyboard, mouse, scanner, microphone).
- Output = results go OUT (monitor, printer, speakers).
- Binary = base 2 = 0 and 1.
- Octal = base 8 = 0–7.
- Decimal = base 10 = 0–9.
- Hexadecimal = base 16 = 0–9 and A–F.
- 8 bits = 1 byte.
- Binary operations = addition, subtraction, multiplication, division.